

Roll No.

Z-9-B

Total No. of Questions : 4]

[Total No. of Printed Pages : 8

11th ARF(SZ)JKUT2024-25

209-B

CHEMISTRY

Time : 3 Hours]

[Maximum Marks : 70

General Instructions :-

- (i) There are total four Sections in the question paper. All questions are compulsory.
- (ii) Section-A contains *ten* Objective Type Questions/Multiple Choice Questions of 1 mark each. (1×10=10)
- (iii) Section-B contains *nine* Very Short Answer Type Questions of 2 marks each to be answered in **20** to **30** words. (2×9=18)
- (iv) Section-C contains *nine* Short Answer Type Questions of 3 marks each to be answered in **100** to **150** words. (3×9=27)
- (v) Section-D contains *three* Long Answer Type Questions of 5 marks each to be answered in **150** to **200** words. (5×3=15)
- (vi) Use log tables, if necessary. Use of scientific calculators is not allowed.

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209-B

Turn Over

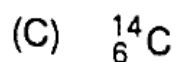
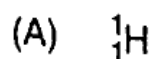
Z-9-B

Section-A

(Objective Type Questions/Multiple Choice Questions) 1 each

1. Select the correct one :

(i) Which of the following is the standard for atomic mass ?

(ii) The number of gram molecules of oxygen in 6.02×10^{24} CO molecules is :

(A) 10 g molecules

(B) 5 g molecules

(C) 1 g molecule

(D) 0.5 g molecule

(iii) The Principle Quantum Number of an electron represents :

(A) Size of the orbital

(B) Spin angular momentum

(C) Orbital angular momentum

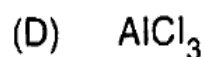
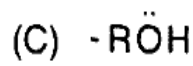
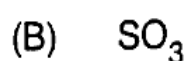
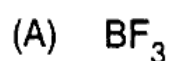
(D) Space orientation of the orbital

(iv) Which has the smallest size ?

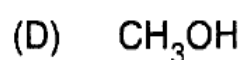
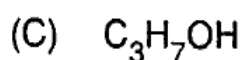
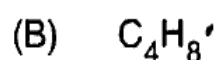
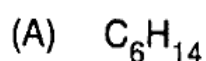


- (v) The triple bond consists of :
- (A) Three sigma bonds
 - (B) Three Pi bonds
 - (C) Two sigma bonds and one Pi bond
 - (D) One sigma and two Pi bonds
- (vi) Atomic orbitals of carbon in CO_2 are :
- (A) sp -hybridised
 - (B) sp^3d hybridised
 - (C) sp^2 hybridised
 - (D) sp^3 hybridised
- (vii) Internal energy of a substance/system is :
- (A) State function
 - (B) Path function
 - (C) Neither state nor path function
 - (D) Both state function as well as path function
- (viii) The oxidation number of iron in Fe_3O_4 is :
- (A) +2
 - (B) +3
 - (C) $8/3$
 - (D) $2/3$

(ix) Which of the following is a nucleophile ?



(x) Which of the following is an unsaturated compound ?



Section-B

(Very Short Answer Type Questions)

2 each

2. Answer the following questions :

(i) What are the main postulates of Dalton's Atomic theory ?

(ii) What do you understand by representative elements ? Name the groups whose elements are called representative elements.

(iii) Explain, why H_2S is a gas while H_2O is liquid, even though both S and O are elements of the same group.

(iv) What do you understand by open and isolated systems ? Give examples.

(v) What are buffer solutions ? How can you say that buffers are used as an important part of many industrial processes ?

- (vi) Explain common ion effect with a suitable example.
- (vii) What are carbanions ? Give examples. What is their order of stability ?
- (viii) What is Wurtz Reaction ? How can it be used to prepare butane ?
- (ix) Discuss briefly the structure of a triple bond.

Section-C

(Short Answer Type Questions)

3 each

3. Answer the following questions :

- (i) N and O combine with H to form NH_3 and H_2O and they combine with each other to form NO_2 . Which law is illustrated ? Explain briefly.
- (ii) What are Quantum Numbers ? Explain the significance of any *two* of the Quantum numbers.
- (iii) What do you mean by Periodic properties of elements ? Explain.
Name any *three* such properties.

- (iv) What is hybridisation ? Explain sp^2 hybridisation with an example.
- (v) State first law of thermodynamics and give a mathematical expression for it. Mention *two* examples to support the law.
- (vi) Define entropy and free energy. How are they related to each other ?
- (vii) What are strong and weak electrolytes ? Derive an expression for the calculation of the degree of ionisation of a weak electrolyte.
- (viii) H_2S acts only as reductant where as SO_2 acts as both oxidant and reductant. Why ? <https://www.jkboseonline.com>
- (ix) Discuss various types of organic reactions (any *three*) with examples.

Section-D

(Long Answer Type Questions)

5 each

4. Answer the following questions :

- (i) State and explain Heisenberg's uncertainty principle.

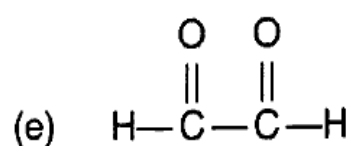
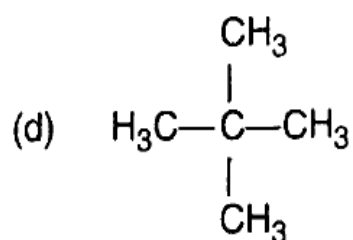
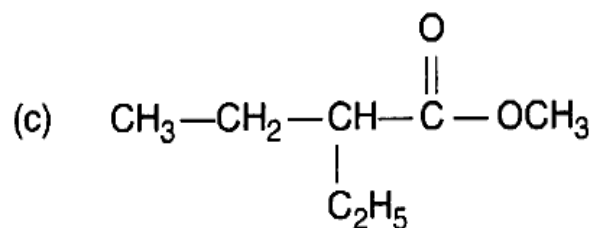
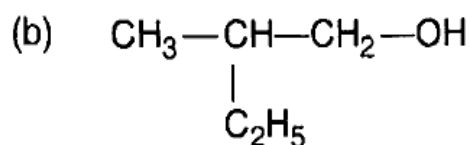
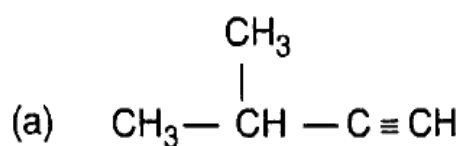
Or

Define an atomic orbital. Discuss the shapes of s , p and d orbitals.

(ii) Explain inductive effect and electromeric effect.

Or

Write IUPAC names of the following compounds :



- (iii) What is Ozonolysis ? How it can be used to determine the position of double bond in an unknown alkene ?

Or

Write short notes on the following :

- (a) Markovnikoff's rule
- (b) Friedel-Crafts acylation
- (c) Benzene resonance